

Chemistry 30 — Unit III

Electrochemical Changes – Lesson 1 Notes

Oxidation and Reduction Reactions

What you will be able to do

In this lesson, you will develop a clear and detailed understanding of oxidation–reduction (redox) reactions. You will learn that redox reactions are defined by the transfer of electrons between reacting species and that oxidation and reduction always occur together.

You will define oxidation and reduction both operationally (in terms of electron loss and gain) and theoretically, identify oxidizing agents and reducing agents, and recognize redox reactions using oxidation numbers and half-reactions. These concepts will be applied to chemical processes that occur in everyday life, including corrosion, as well as in living systems such as cellular respiration and photosynthesis.

This lesson forms the conceptual foundation for the entire electrochemistry unit. A strong understanding of oxidation and reduction is essential before studying voltaic cells, electrolytic cells, standard reduction potentials, and redox titrations.

PART A — Foundations of Oxidation–Reduction Reactions

Oxidation–reduction reactions form the conceptual foundation of electrochemistry. They explain how electrons are transferred during chemical reactions and why some reactions occur spontaneously while others do not.

In everyday language, oxidation is often associated with oxygen and reduction with the removal of oxygen. These definitions originated from early observations of metals reacting with oxygen in air. For example, iron reacts slowly with oxygen to form rust, while metal ores were heated to remove oxygen and produce pure metals.

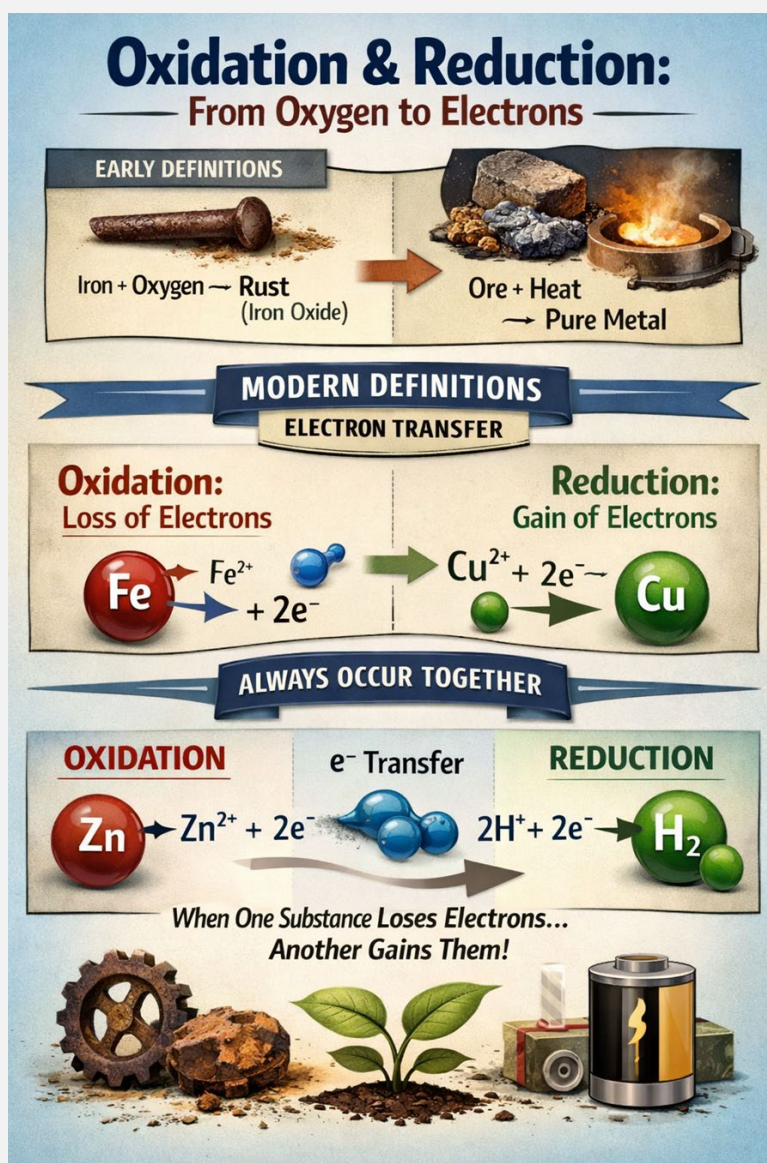
As chemistry developed, scientists recognized that many reactions displayed the same characteristics even when oxygen was not involved. To account for these observations, oxidation and reduction were redefined in terms of electron transfer.

According to modern chemical theory:

- Oxidation is the loss of electrons.
- Reduction is the gain of electrons.

Because electrons cannot be created or destroyed in a chemical reaction, oxidation and reduction must always occur together. When one substance loses electrons, another substance must gain them. For this reason, oxidation and reduction are inseparable processes.

Understanding oxidation and reduction as electron-transfer processes allows chemists to explain corrosion, metal reactivity, biological energy transfer, and the operation of electrochemical cells. This electron-based definition will be used throughout the remainder of this unit.



PART B — Operational Definitions of Oxidation and Reduction

To analyze oxidation–reduction reactions consistently, chemists use operational definitions based on observable electron transfer. These definitions allow reactions to be classified and compared, even when oxygen is not involved.

In operational terms, oxidation and reduction are defined by the movement of electrons. These definitions apply to all redox reactions, regardless of the substances involved.

Oxidation is defined as the loss of one or more electrons by a substance. When a species is oxidized, its electrons are transferred to another species. As a result, the oxidized substance becomes more positively charged or less negatively charged.

Reduction is defined as the gain of one or more electrons by a substance. When a species is reduced, it accepts electrons from another species. As a result, the reduced substance becomes more negatively charged or less positively charged.

Because electrons are conserved in chemical reactions, oxidation and reduction must always occur together. A substance cannot lose electrons unless another substance gains them. For this reason, oxidation and reduction are described as complementary processes within a single redox reaction.

For example, when zinc metal reacts with aqueous copper(II) ions, zinc atoms lose electrons and are oxidized to zinc ions, while copper(II) ions gain electrons and are reduced to copper metal. Although these changes are described separately, they occur simultaneously.

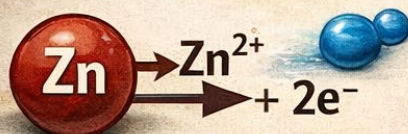
Operational definitions of oxidation and reduction will be used throughout this unit to identify redox reactions, determine the direction of electron flow, and analyze electrochemical cells.

Operational Definitions of Oxidation & Reduction

Movement of Electrons in Redox Reactions

OXIDATION

Loss of Electrons



↑ More Positively Charged
Becomes Less Negatively Charged

REDUCTION

Gain of Electrons



↓ Less Positively Charged
Becomes More Negatively Charged

ALWAYS OCCUR TOGETHER



More Positively Charged



Less Positively Charged

Electrons Lost
By One Substance...



...Are Gained
By Another!



Change Occurs Simultaneously!

PART C — Oxidizing and Reducing Agents

In every oxidation–reduction reaction, electrons are transferred from one substance to another. The substances responsible for accepting and donating electrons are known as oxidizing agents and reducing agents. Identifying these agents is essential for understanding and predicting redox reactions.

An oxidizing agent is a substance that causes another substance to be oxidized. An oxidizing agent accomplishes this by gaining electrons from another species. Because it gains electrons, the oxidizing agent itself is reduced during the reaction.

A reducing agent is a substance that causes another substance to be reduced. A reducing agent accomplishes this by losing electrons to another species. Because it loses electrons, the reducing agent itself is oxidized during the reaction.

It is important to distinguish between the process and the agent involved. Oxidation and reduction describe what happens to a substance, while oxidizing agents and reducing agents describe the role a substance plays in causing those changes.

For example, consider the reaction between zinc metal and aqueous copper(II) ions:

- Zinc metal loses electrons and is oxidized. Zinc is therefore the reducing agent.
- Copper(II) ions gain electrons and are reduced. Copper(II) ions are therefore the oxidizing agent.

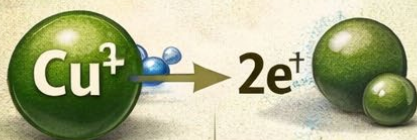
In general, metals tend to act as reducing agents because they readily lose electrons, while nonmetals and metal ions tend to act as oxidizing agents because they readily gain electrons. These tendencies will be explored in greater detail later in this unit using redox tables and standard reduction potentials.

Oxidizing Agents & Reducing Agents

Roles in Redox Reactions

OXIDIZING AGENT

GAINING ELECTRONS



GAINING
ELECTRONS

IS REDUCED

Causes Other
to be Oxidized

Copper(II) Ions
(Cu^{2+})

REDUCING AGENT

LOSING ELECTRONS



LOSING
ELECTRONS

IS OXIDIZED

Reducing
Agents

Zinc Metal
(Zn)



List of Oxidizing Agents Levels Out List of Reducing Agents

PART D — Identifying Oxidation and Reduction Using Oxidation Numbers

Oxidation numbers provide a systematic method for tracking electron transfer in oxidation–reduction reactions. By examining changes in oxidation numbers, chemists can determine which substances are oxidized and reduced, even in complex reactions.

An oxidation number (also called an oxidation state) is an assigned value that represents the hypothetical charge an atom would have if all bonds to other atoms were considered to be ionic. Oxidation numbers are not actual charges but are useful bookkeeping tools for tracking electron transfer.

The following rules are used to assign oxidation numbers:

- The oxidation number of an element in its elemental form is 0.
- The oxidation number of a monatomic ion is equal to its ionic charge.
- Oxygen usually has an oxidation number of -2 .
- Hydrogen usually has an oxidation number of $+1$ when bonded to nonmetals.
- The sum of oxidation numbers in a neutral compound is 0.
- The sum of oxidation numbers in a polyatomic ion equals the charge on the ion.

In a redox reaction, oxidation is identified by an increase in oxidation number, while reduction is identified by a decrease in oxidation number. These changes reflect the loss or gain of electrons.

For example, when zinc metal reacts with copper(II) ions, the oxidation number of zinc increases from 0 to $+2$, indicating oxidation. The oxidation number of copper decreases from $+2$ to 0, indicating reduction.

Oxidation numbers will be used extensively in this unit to identify redox reactions, determine oxidizing and reducing agents, and analyze electron transfer in electrochemical cells.

Rules for Assigning Oxidation Numbers

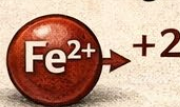
Tracking Electron Transfer in Redox Reactions

GENERAL RULES

1 Elemental Form
Oxidation Number = 0



2 Monatomic Ion:
Oxidation Number = Ionic Charge



3 Oxygen:
Usually = -2



5 Sum in Compounds = 0
Sum in Polyatomic Ion = Charge of Ion

5 Sum in Polyatomic Ion = Charge of Ion

OXIDATION

Increase



Loss of Electrons
More Positively Charged

Redox Reactions
Involve:

Change in Oxidation
Numbers

REDUCTION

Decrease



Gain of Electrons
More Negatively Charged



OXIDATION

Increase
Loss of Electrons

Zinc Metal is Oxidized



REDUCTION

Decrease
Gain of Electrons

Copper(II) Ions are Reduced



Use Oxidation Numbers to
Redox Reactions!

Track Electron Flow in
Redox Reactions!



PART E — Recognizing Oxidation–Reduction (Redox) Reactions

Not all chemical reactions involve electron transfer. Being able to recognize whether a reaction is a redox reaction is an essential skill before writing half-reactions or analyzing electrochemical cells.

A chemical reaction is classified as an oxidation–reduction reaction if electrons are transferred between reacting species. This electron transfer can be identified by changes in oxidation numbers.

If at least one element shows an increase in oxidation number and another element shows a decrease in oxidation number, a redox reaction has occurred. Both changes must be present because oxidation and reduction always occur together.

Many common reaction types are redox reactions. Single replacement reactions are always redox reactions because one element replaces another by transferring electrons. Combustion reactions are also redox reactions, as electrons are transferred to oxygen.

In contrast, double replacement reactions and acid–base neutralization reactions generally do not involve electron transfer. In these reactions, oxidation numbers remain unchanged, and therefore no redox process occurs.

For example, when aqueous sodium chloride reacts with aqueous silver nitrate to form solid silver chloride, the oxidation numbers of all elements remain the same. This reaction is not a redox reaction.

Being able to distinguish redox reactions from non-redox reactions will be critical in later lessons, particularly when writing half-reactions and analyzing electrochemical cells.

Identifying Redox Reactions

When Electrons Are Transferred During a Reaction

REDOX REACTIONS

OXIDATION NUMBERS CHANGE



OXIDATION

Increase Zinc: $0 \rightarrow +2$

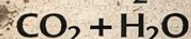
Single Replacement Reactions



REDUCTION

Decrease Copper: $+2 \rightarrow 0$

Combustion Reactions

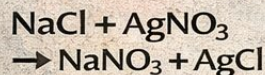


NOT REDOX REACTIONS

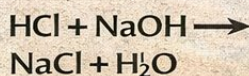
OXIDATION NUMBERS REMAIN THE SAME



Double Replacement Reactions



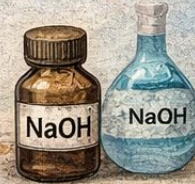
Acid-Base Neutralization Reactions



No Change in Oxidation Numbers
No Redox Process Occurs!



Zinc Metal is Oxidized



Copper(II) Ions are Reduced

Lesson 1 Summary

In this lesson, you learned that oxidation–reduction (redox) reactions are defined by the transfer of electrons between substances. Oxidation is the loss of electrons, while reduction is the gain of electrons, and these processes must always occur together.

You learned how to identify oxidizing agents and reducing agents, how oxidation numbers are used to track electron transfer, and how to recognize whether a chemical reaction is a redox reaction. These ideas form the foundation of electrochemistry and are essential for understanding electrochemical cells and electron flow.

Common Misconceptions

- Oxidation always involves oxygen. (False — oxidation refers to electron loss.)
- Reduction means decreasing mass or size. (False — reduction refers to electron gain.)
- The oxidizing agent is oxidized. (False — the oxidizing agent is reduced.)
- Oxidation and reduction can occur separately. (False — they always occur together.)
- All reactions are redox reactions. (False — many double replacement and acid–base reactions are not.)

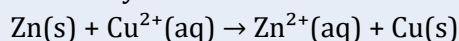
Looking Ahead to Lesson 2 — Half-Reactions

In the next lesson, you will learn how oxidation and reduction processes can be separated into half-reactions. Half-reactions show explicitly how electrons are transferred and provide a powerful tool for analyzing redox reactions quantitatively.

You will learn how to write and balance half-reactions and how they are combined to represent overall redox processes. These skills are essential for understanding electrochemical cells and calculating cell potentials.

Review Questions — Lesson 1

1. Identify which substance is oxidized and which is reduced in the reaction:



Solution:

Zinc loses electrons and its oxidation number increases from 0 to +2, so zinc is oxidized.

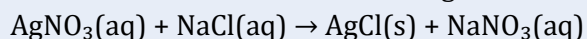
Copper(II) ions gain electrons and their oxidation number decreases from +2 to 0, so copper is reduced.

2. Identify the oxidizing agent and reducing agent in the reaction above.

Solution:

Zinc is the reducing agent because it loses electrons. Copper(II) ions are the oxidizing agent because they gain electrons.

3. Determine whether the following reaction is a redox reaction:



Solution:

The oxidation numbers of all elements remain unchanged, so no electrons are transferred. This reaction is not a redox reaction.

4. State the oxidation numbers of sulfur in SO_2 and SO_4^{2-} and determine which process occurs.

Solution:

In SO_2 , sulfur has an oxidation number of +4. In SO_4^{2-} , sulfur has an oxidation number of +6. Because the oxidation number increases, sulfur is oxidized.

5. Explain why oxidation and reduction must always occur together.

Solution:

Electrons cannot be created or destroyed in a chemical reaction. If one substance loses electrons, another substance must gain them. Therefore, oxidation and reduction must occur simultaneously.